Uncertainty and failure analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Uncertainty and failure analysis: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Distinguish a predicted class, **confidence score** and observed correctness

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a reused feature extractor and a newly fitted **output** head

Reference example, not the only acceptable wording: Teacher explanation: The extractor gives h = 6 and the head predicts 1.

Weights [1, 1] and bias 0 stay frozen; only the threshold is fitted. h is a nonnegative sum of counts, with no probability mapping or **calibration**.

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Look for: Look up a vector using a token ID and preserve sequence shape

Reference example, not the only acceptable wording: Teacher explanation: Vectors are [[1, 1], [0, 1], [0, 2], [0, 0]], mask [1, 1, 1, 0] and shape 4 by 2.

Training a small image model

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a complete image model with fixed features and one trained weight

Reference example, not the only acceptable wording: Teacher explanation: d = 2, h1 = 2 and h2 = 0. Score is 0.9, **prediction** class 1 and squared error (0.9 - 1) squared = 0.01.

For the compact alternative schedule, use two 45-minute sessions. These are supplied classifier **output**s for arithmetic and reasoning, not calibrated probabilities from a trained model. Keep overall accuracy, accepted-case accuracy and **coverage** visible together.

Do not imply that review guarantees correction or that model confidence diagnoses the cause of uncertainty.

Outcomes

Distinguish a predicted class, **confidence score** and observed correctness

Calculate accuracy and **coverage** for a review policy

Investigate failure patterns without claiming an untested cause

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Two constructed cases provide very limited **evidence**. New **input**s can differ, and validation used to choose settings is not untouched final evaluation. Performance needs representative independent testing.

End goal for the pictorial case

By the end, I can explain uncertainty and failure analysis using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Uncertainty and failure analysis using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Uncertainty and failure analysis · 1

A model can be confidently wrong. Use these supplied **output**s for a fictional image classifier: class 1 means a triangle is present; class 0 means it is absent.

The values p1 are probability estimates for class 1, invented for this exercise rather than produced by a trained model. Predict 1 when p1 >= 0.5 and 0 otherwise. The selected-class confidence is max(p1, 1 - p1).

A predicts 1 with confidence 0.95 and is correct. F predicts 0 with confidence 0.90 and is wrong: p1 = 0.10 is not low confidence in the selected class.

Reference labels are supplied for this exercise; real investigations must check their quality too.

A score alone cannot establish reliability. **Calibration** asks whether, across many comparable cases, **prediction**s assigned a probability such as 0.90 are correct about 90 percent of the time.

One correct or incorrect case cannot answer that question.

Explain Uncertainty and failure analysis · 2

The **prediction**s for A through F are 1, 1, 1, 1, 0, 0. A, C and E are correct, so overall accuracy is 3 / 6 = 50 percent.

There are two false positives (B and D), one false negative (F), two true positives and one true negative. Different errors can matter differently; a single accuracy number hides that distinction.

Explain Uncertainty and failure analysis · 3

The function returns overall accuracy, **coverage** and accuracy among accepted cases. It assumes numeric scores and binary labels. None means no cases were accepted, so accepted-case accuracy is undefined, not zero or perfect.

Keep counts with proportions and declare the comparison at the boundary.

Term	Meaning
Confidence score	The model’s score for its selected class, which may not match observed reliability
Calibration	Agreement between predicted probabilities and observed frequencies across many comparable cases
Abstention	Withholding an automatic prediction and following a declared alternative process
Coverage	The fraction of all cases on which a system makes an automatic prediction

Confidence does not remove every error

CASES	AT CUTOFF 0.80	OBSERVED RESULT
A, B, C, F	Accept automatically	2 correct, 2 wrong
D, E	Defer for review	1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Explain Uncertainty and failure analysis · 4

Guided: only A is accepted. **Coverage** is 1 of 6; accepted accuracy is 1 of 1. This single constructed success, selected by a rule examined on these data, does not establish future reliability.

Exit: report one correct automatic **prediction** out of one accepted, with five of six deferred and no reviewer outcomes.

Require representative untouched evaluation, subgroup and **calibration evidence**, measured reviewer outcomes and workload, and a comparison with a declared baseline.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Confidence does not remove every error

CASES	AT CUTOFF 0.80	OBSERVED RESULT
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Model the reasoning aloud · 5

Deferred cases are not automatically repaired. A review workflow needs an assigned reviewer, suitable **evidence**, a decision record and an outcome check.

Report review workload and the final system’s results separately; we have no reviewer outcomes here. Do not erase deferred cases from the report or count them as correct.

Raise the cutoff to 0.95 for the same six cases. Which cases are accepted, what are coverage and accepted-case accuracy, and why does that result not guarantee perfect future predictions?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Uncertainty and failure analysis

Keep positive-class probability separate from confidence in the chosen label.

The facts we will use

Supplied facts

Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/29

Positive probability is not always chosen-label confidence

	p(class 1)	prediction	confidence	reference
A	0.9	1	0.9	0
B	0.6	1	0.6	1
C	0.2	0	0.8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/29

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.



3. Selective review

Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predictions	A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.
2. Selected-label confidence	Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.
3. Selective review	Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

What this example does not establish

Keep the stated limit

Supplied scores are not proven calibrated. Review policies need representative held-out

evidence and clear handling of rejected cases.

Change one thing in the pictured case

Change one thing

Change: Raise the acceptance cutoff to 0.85.

Predicted result

Prediction: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Why the result follows

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Uncertainty and failure analysis, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the limits of the two-case validation result from Week 28. Define the triangle-present label.
7 to 18 minutes	Trace A and F, then complete W1 with selected-class confidence distinct from p1.
18 to 32 minutes	Build the error tally and complete W2. Trace the report function and its three returned quantities.
32 to 45 minutes	Use the visual and W3 to retain deferred cases in workload reporting and handle an empty accepted set.
Session 2 start to 8 minutes	Complete the guided cutoff 0.95 case and critique a perfect-accuracy claim.
8 to 20 minutes	Inspect the blur pattern and design a controlled comparison with source-grouped splits.
20 to 35 minutes	Collect W4 independently, including denominators and the worse accepted-subset result.
35 to 45 minutes	Use W5 and the exit response to evaluate causal and reliability claims. Give feedback, then the fresh retry.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate boxes for p1, predicted class, confidence, reference label and correctness. Use fractions before percentages. Extension: for A, B, C and F, mean confidence is 0.8875 while observed accuracy is 0.5.

Discuss why this small grouped discrepancy motivates **calibration** study but cannot establish a population-wide **calibration** rate.

Reduce help deliberately

Completion practice

Completion task: For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Reasoning to listen for

Teacher check: Teacher explanation: False positives are B and D; the false negative is F. A and C are the two true positives; E is the one true negative. Accuracy is 3 of 6, or 50 percent.

It hides which errors occur and how their consequences differ.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p1 always means low confidence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: For a binary model, p1 = 0.10 selects class 0 with confidence 0.90.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

p(class 1)=0.2; predicted class 0 has selected-label confidence 0.8. These quantities answer different questions. Changed-case check: Only A is accepted: accuracy 0/1, **coverage** 1/3.

The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Changed-case reasoning: Only A is accepted: accuracy 0/1, **coverage** 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Uncertainty and failure analysis investigation and use the first unsupported step to choose a teaching response.

A low p1 always means low confidence: For a binary model, p1 = 0.10 selects class 0 with confidence 0.90.

Reviewing uncertain cases fixes all errors: B and F remain confidently wrong; deferred outcomes also need measurement.

A shared condition proves the cause: A small error pattern motivates investigation; label checks, controlled examples and independent evaluation are still needed.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider **coverage**.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Uncertainty and failure analysis questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: only A is accepted. **Coverage** is 1 of 6; accepted accuracy is 1 of 1. This single constructed success, selected by a rule examined on these data, does not establish future reliability.

Exit: report one correct automatic **prediction** out of one accepted, with five of six deferred and no reviewer outcomes.

Require representative untouched evaluation, subgroup and **calibration evidence**, measured reviewer outcomes and workload, and a comparison with a declared baseline.

W1 Separate confidence from correctness

Workbook question · W1

For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

Teacher reasoning · W1

Teacher explanation: B predicts 1 with confidence 0.90 and is wrong. E predicts 0 with confidence 0.55 and is correct. F predicts 0 with confidence 0.90 and is wrong.

F's low score for class 1 corresponds to a high score for class 0, the class selected.

W2 Count both types of error

Workbook question · W2

For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Teacher reasoning · W2

Teacher explanation: False positives are B and D; the false negative is F. A and C are the two true positives; E is the one true negative. Accuracy is 3 of 6, or 50 percent.

It hides which errors occur and how their consequences differ.

W3 Audit the review policy

Workbook question · W3

At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?

What does the report function return for accepted-case accuracy at cutoff 0.99?

Teacher reasoning · W3

Teacher explanation: Accepted cases are A, B, C, F; deferred cases are D and E. **Coverage** is 4 of 6 and accepted accuracy is 2 of 4. No reviewer outcomes are supplied, so final system accuracy is unknown.

At 0.99 no case is accepted; accepted accuracy is undefined and the function returns None.

W4 Analyse new outputs independently

Workbook question · W4

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

Teacher reasoning · W4

Teacher explanation: **Prediction**s are 1, 1, 0, 0; confidences are 0.85, 0.70, 0.70, 0.95. L, M and N are correct, so overall accuracy is 3 of 4. K and N are accepted; **coverage** is 2 of 4 and accepted accuracy is 1 of 2.

Accepted accuracy is lower than overall accuracy in this example.

W5 Design a failure investigation

Workbook question · W5

All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

Identify the unsupported claim and the split problem; propose a better investigation.

Teacher reasoning · W5

Teacher explanation: Three failures show a pattern, not a proven cause. Related clear and blurred copies across splits can leak source-image information.

Check labels and pre**process**ing, then hold related versions together in reserved source groups and compare controlled blur variants within those groups.

Choose changes before new final evaluation and report the small sample limitation.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires **prediction**s 1, 1, 0, 0; confidences 0.85, 0.70, 0.70, 0.95; accuracy 3 of 4; accepted K and N, **coverage** 2 of 4 and accepted accuracy 1 of 2.

Fresh retry: (p1, label) values (0.90, 1), (0.65, 0), (0.35, 1), (0.10, 0). **Prediction**s are 1, 1, 0, 0; confidences 0.90, 0.65, 0.65, 0.90. Overall accuracy is 2 of 4.

At cutoff 0.80 accept first and fourth: **coverage** 2 of 4 and accepted accuracy 2 of 2. Require a limitation despite the perfect accepted subset.

Assess independent transfer

Independent question · withhold the key

Selected case: Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: **Prediction**s are 1, 1, 0, 0; confidences are 0.85, 0.70, 0.70, 0.95. L, M and N are correct, so overall accuracy is 3 of 4. K and N are accepted; **coverage** is 2 of 4 and accepted accuracy is 1 of 2.

Accepted accuracy is lower than overall accuracy in this example.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Uncertainty and failure analysis to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled uncertainty and failure analysis **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, **confidence score** and observed correctness.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate boxes for p1, predicted class, confidence, reference label and correctness. Use fractions before percentages. Extension: for A, B, C and F, mean confidence is 0.8875 while observed accuracy is 0.5.

Discuss why this small grouped discrepancy motivates **calibration** study but cannot establish a population-wide **calibration** rate.

Week 30 studies dataset shift, comparing development and new-use conditions without assuming a familiar **confidence score** implies familiar data.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/30, ai-5/32, ai-5/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Uncertainty and failure analysis lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Distinguish a predicted class, **confidence score** and observed correctness

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy.

Why might a single accuracy hide useful information?

Supported practice answer

Reasoned practice answer: Teacher explanation: False positives are B and D; the false negative is F. A and C are the two true positives; E is the one true negative. Accuracy is 3 of 6, or 50 percent.

It hides which errors occur and how their consequences differ.

Check every learner before discussion: Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p1 always means low confidence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: For a binary model, p1 = 0.10 selects class 0 with confidence 0.90.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Raise the acceptance cutoff to 0.85. Require a **prediction** and a reason before comparing. Expected result: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

TEACHER GUIDE · 9 OF 9

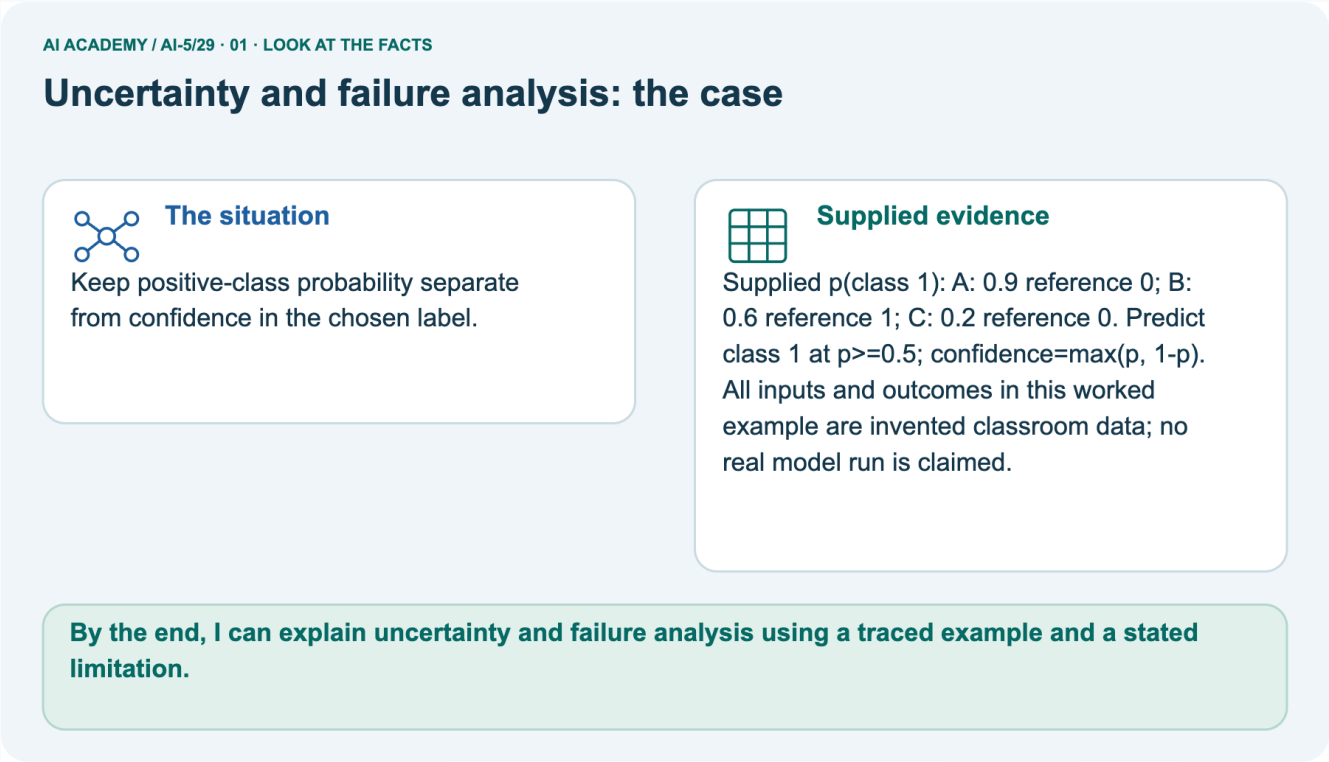
Visual concept journey

> Show one Uncertainty and failure analysis graphic at a time, request a prediction and discuss

> its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Uncertainty and failure analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep positive-class probability separate from confidence in the chosen label. Supplied facts: Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

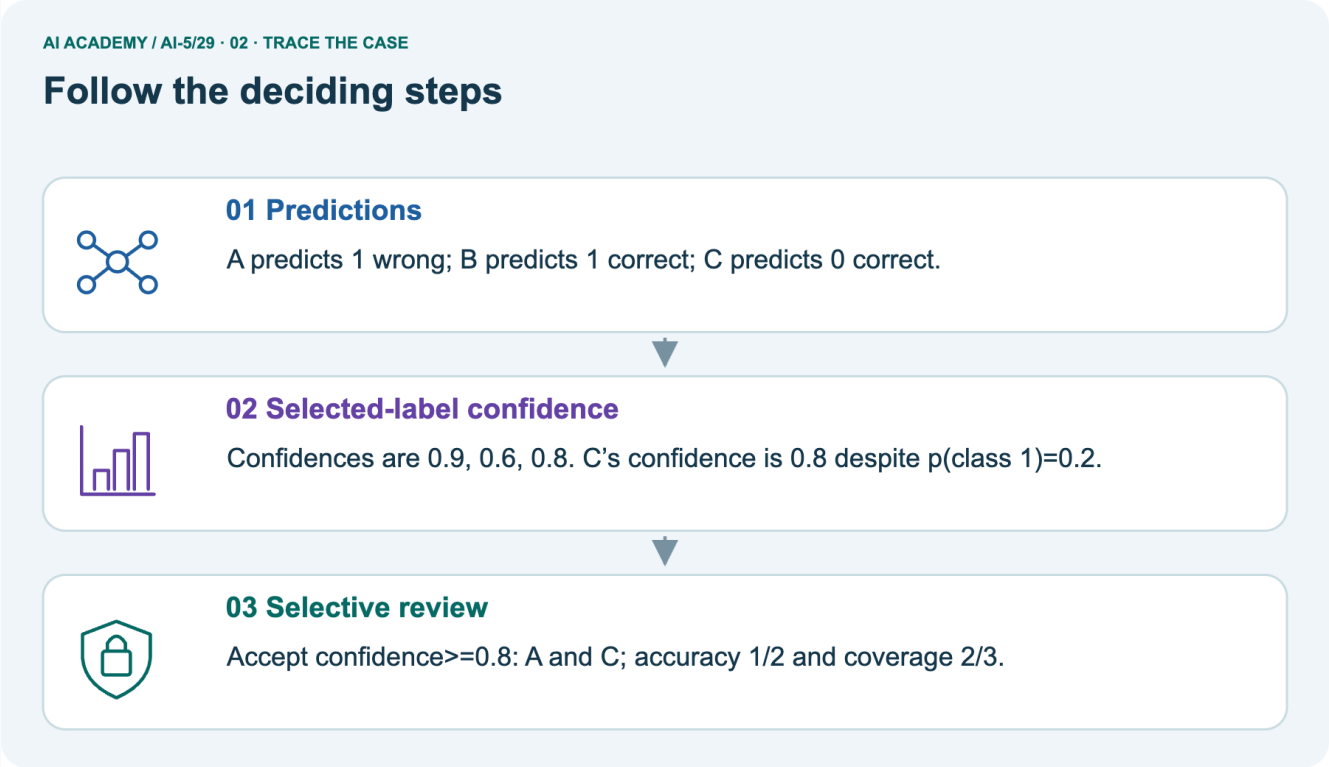
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Predictions: A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct. → Selected-label confidence: Confidences are 0.9, 0.6, 0.8.

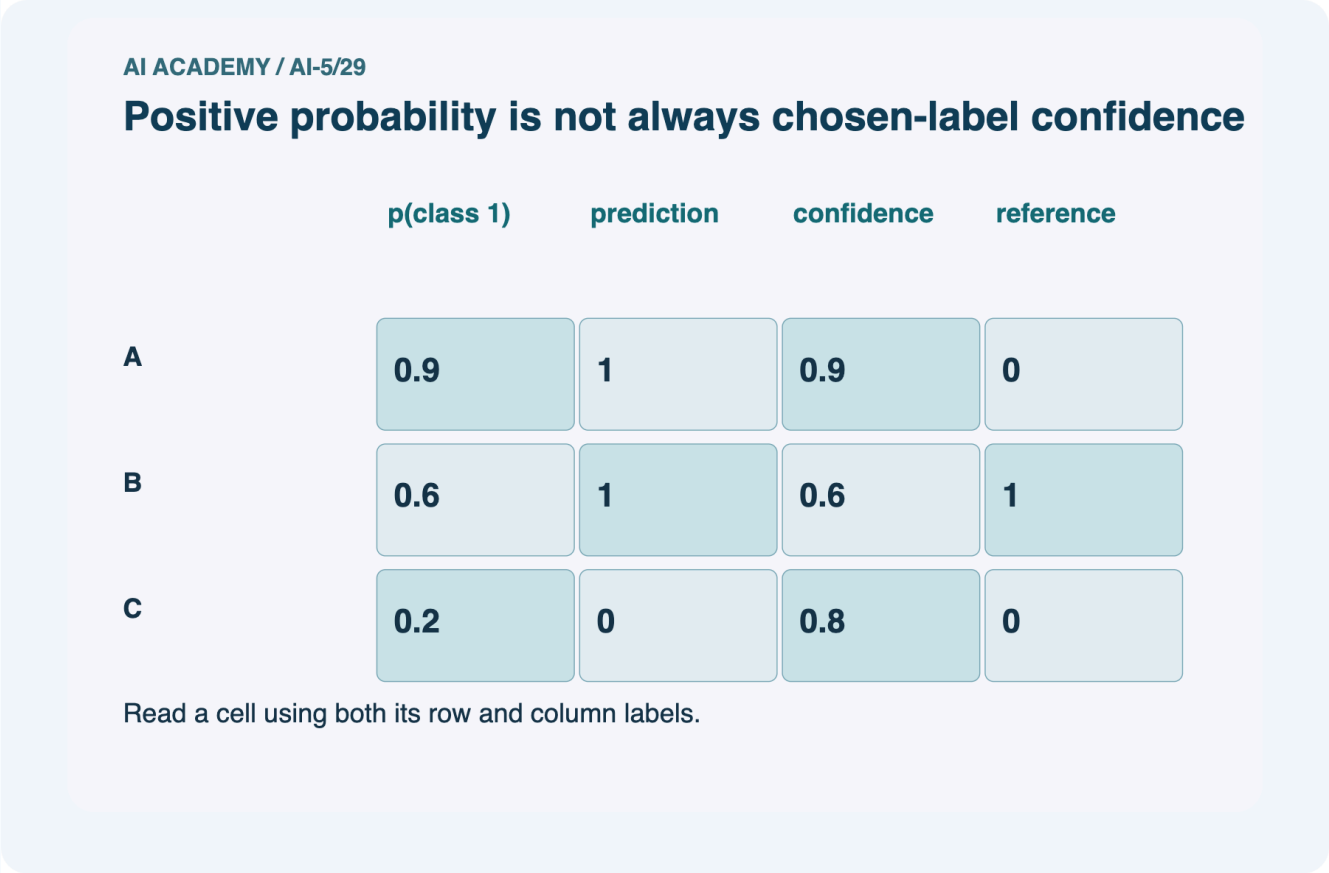
C's confidence is 0.8 despite p(class 1)=0.2. → Selective review: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Positive probability is not always chosen-label confidence



Follow the labelled picture with the matching explanation. Open the vector diagram

Positive probability is not always chosen-label confidence. A: p(class 1) = 0.9, **prediction** = 1, confidence = 0.9, reference = 0. B: p(class 1) = 0.6, **prediction** = 1, confidence = 0.6, reference = 1.

C: p(class 1) = 0.2, **prediction** = 0, confidence = 0.8, reference = 0.

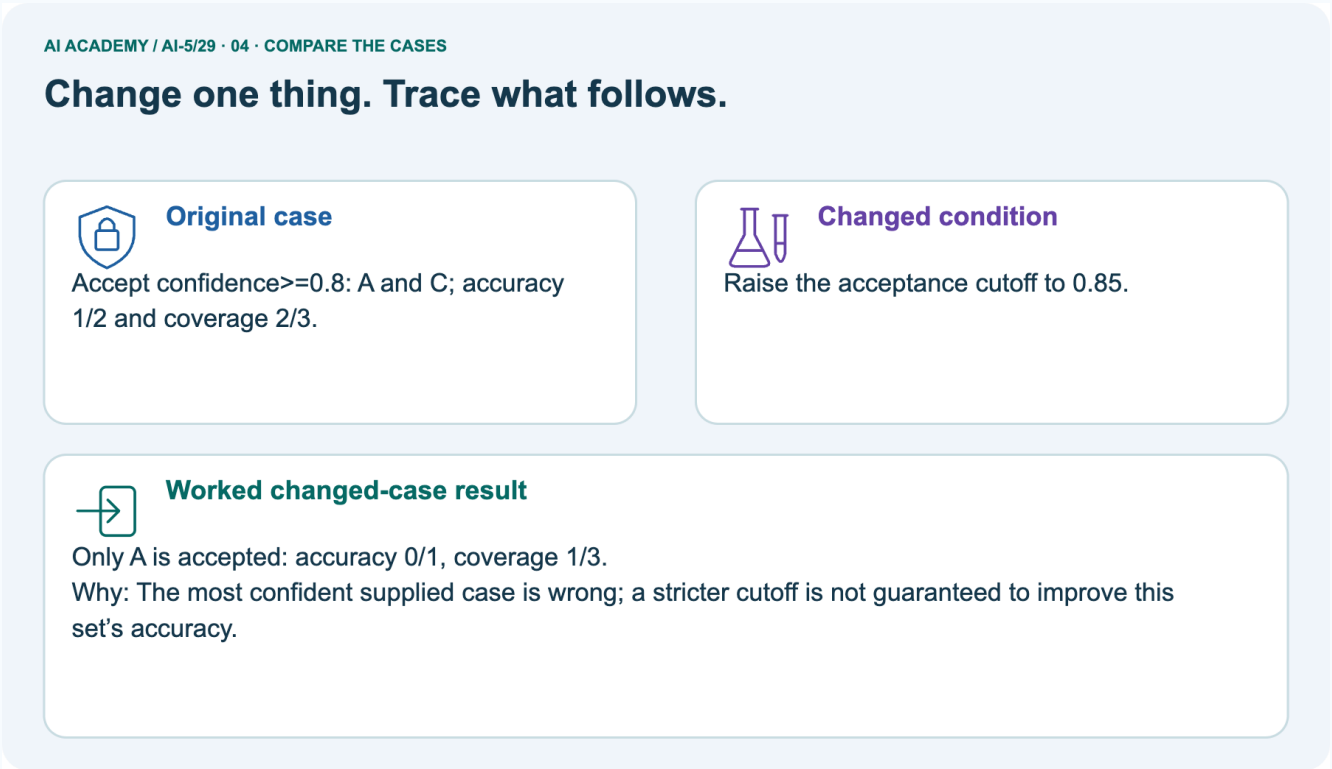
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3. Changed condition: Raise the acceptance cutoff to 0.85. Expected result: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set’s accuracy.

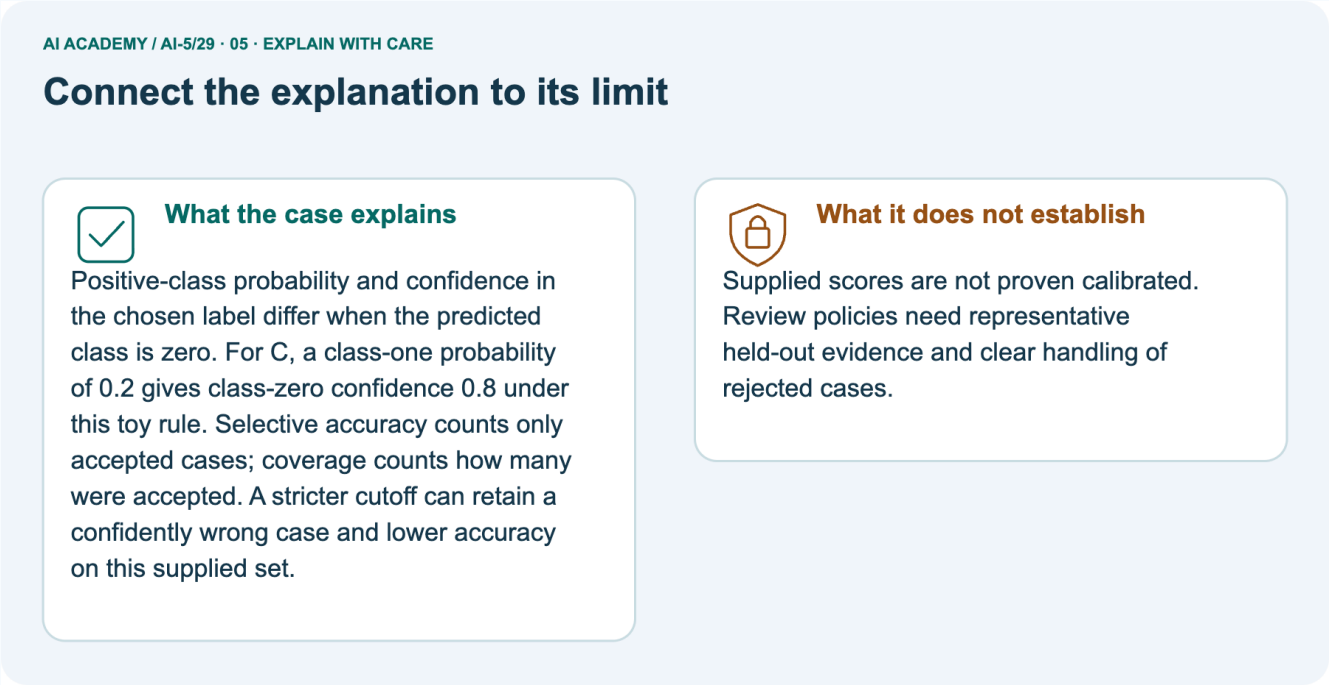
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set’s accuracy.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Limit: Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

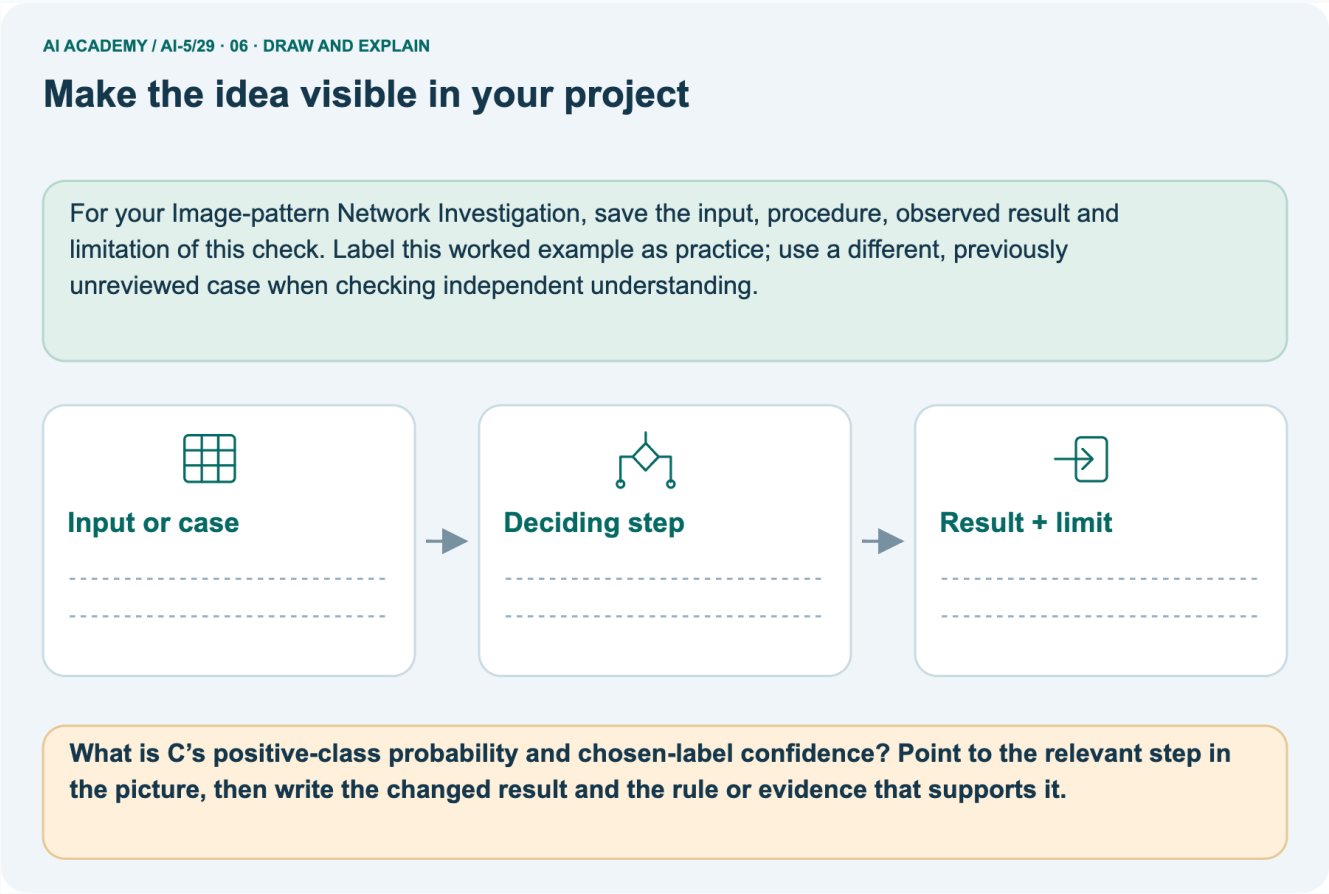
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: p(class 1)=0.2; predicted class 0 has selected-label confidence 0.8. These quantities answer different questions. Changed-case check: Only A is accepted: accuracy 0/1, **coverage** 1/3.

The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set’s accuracy.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.